

Hector Hernandez

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Summary

ML/Agentic AI Engineer with 10 years in Intel production environments and hands-on experience building LLM-based tools, multi-agent systems, and scalable data platforms. I design reliable agentic systems that combine models, tools, and data into end-to-end multi-step workflows with explicit planning, structured outputs, tool use, failure handling, and monitoring and observability. Currently completing an MS in Computer Science at ASU while applying LLM-assisted coding, evaluation, and latency/cost/accuracy tradeoffs to ship practical AI systems.

Education

Arizona State University

Expected December 2026

MS in Computer Science (GPA: 3.88/4.0)

- **Coursework:** Agentic AI, Statistical Machine Learning, Artificial Intelligence, Data Processing at Scale, Advanced Operating Systems, Data Structures and Algorithms, Applied Cryptography, Data Visualization.

Arizona State University

Aug 2012 – May 2015

BS in Chemical Engineering

Experience

Data Engineering Intern

Atlanta, GA

Honeywell

May 2026 – August 2026

- Building Python and SQL data pipelines that support analytics and reporting workflows across business units, with emphasis on scalable data models and downstream reliability.
- Collaborating with engineering teams to improve data quality, automate manual processing, and ship maintainable backend systems.

Software Application Engineer

Chandler, AZ

Intel

June 2020 – July 2025

- Designed end-to-end architecture for Intel's global EDI manufacturing data platform, integrating Python prototypes, Angular/D3/Highcharts visualizations, and backend services for thousands of engineering users.
- Re-architected a legacy Python/Oracle ETL stack into C#/SQL Server, making cost/performance tradeoffs that improved scalability, maintainability, and long-term operational efficiency.
- Optimized database structures and query paths across services, reducing data transfer latency from 30+ seconds to under 3 seconds for production engineering workflows.
- Built Airflow, Docker, Python, and SQL Server pipelines with clear failure handling and reduced manual intervention, improving reliability of recurring manufacturing data workloads.
- Maintained globally distributed production infrastructure, resolving bugs and enhancements while strengthening monitoring, observability, and availability for manufacturing teams.
- Led code reviews and Python standards for the EDI development team, using LLM-assisted coding and structured review habits to improve quality, testability, and defect prevention.

Photolithography Process Engineer

Chandler, AZ

Intel

June 2015 – May 2020

- Sustained and improved Photolithography processes to meet manufacturing demand for throughput, cycle time, and yield.
- Improved quality levels by reducing scrap, minimizing yield loss mechanisms, and enhancing process control.
- Identified and executed cost-reduction solutions aligned with factory budget goals.
- Collaborated with Manufacturing, Maintenance, and Process teams to optimize the overall photolithography area, working extensively with both Nikon and ASML scanners.
- Reduced defectivity mechanisms through targeted engineering solutions and tighter process control.
- Developed and deployed automation tools, including 1-click reports, SQL queries, JSL scripts, and Python scripts to extract, process, and visualize large datasets from legacy and modern systems.
- Trained and mentored multiple new engineers, providing hands-on guidance to help them understand semiconductor equipment and processes, accelerating their ramp-up in the photolithography area.

AI Projects & Research

VoiceValet

[GitHub](#) 

- Built an agentic phone agent using Claude, Vapi, Deepgram, and Resend to execute outbound calls, integrate tools, handle multi-step tasks, recover from failures, and return transcript summaries by email.

ConvergeAI

[GitHub](#) 

- Built a multi-model consensus solver that runs GPT and Claude in parallel, compares disagreement, refines answers, caches responses, and tracks cost for latency/cost/accuracy tradeoffs.

Claude MCP Server Collection

[GitHub](#) 

- Created 5 MCP servers for stocks, weather, news, Gmail, and PostgreSQL finance data, extending Claude with real-time API access, read-only queries, OAuth actions, and graceful error handling.

Nexus Research

[GitHub](#) 

- Built a multi-agent research pipeline that decomposes questions, retrieves public API context, checks reliability, and emits structured Markdown reports.

τ -bench Research

CSE 598 Research

- Evaluated agentic tool use across Qwen3 and Gemma4 families using τ -bench tasks, error taxonomies, $pass^k$ metrics, vLLM, SLURM, and HPC.

Technologies

Agentic AI & LLMs: Claude API, OpenAI API, MCP, LangChain, Ollama, vLLM, ReAct, Function Calling, multi-agent orchestration, tool-use evaluation

Backend & Systems: Python, FastAPI, Docker, PostgreSQL, SQL Server, Airflow, C#, .NET

ML & Evaluation: scikit-learn, Pandas, NumPy, τ -bench, WildToolBench, $pass^k$ metrics

Dev & Infra: Git, GitHub, Claude Code CLI, SLURM, HPC (A100 / AMD ROCm), VS Code

Supporting: JavaScript, TypeScript, React, Angular, D3.js